Lecture 5: Decidability and Undecidability

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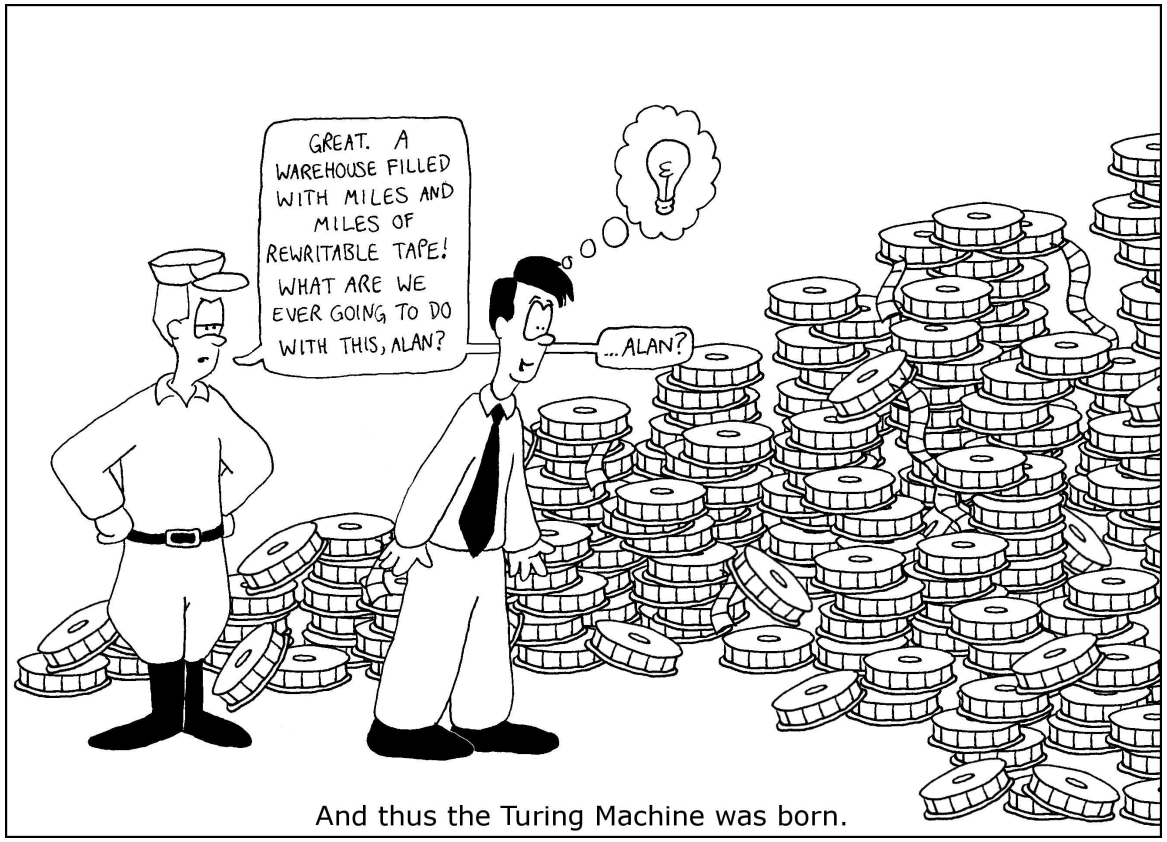
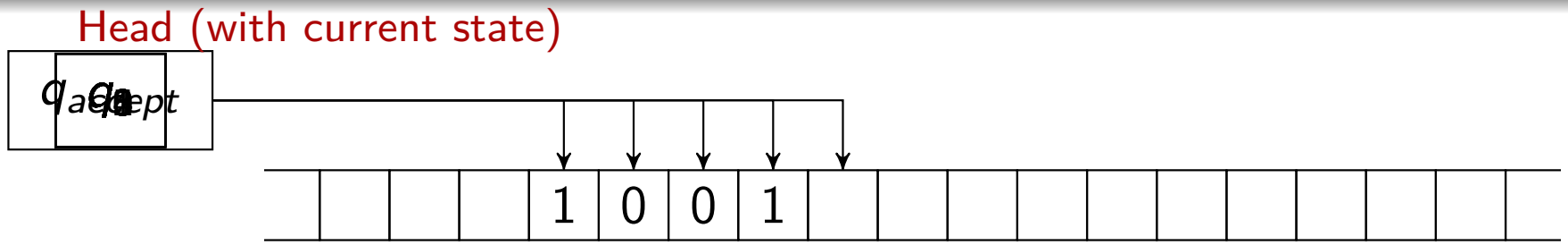
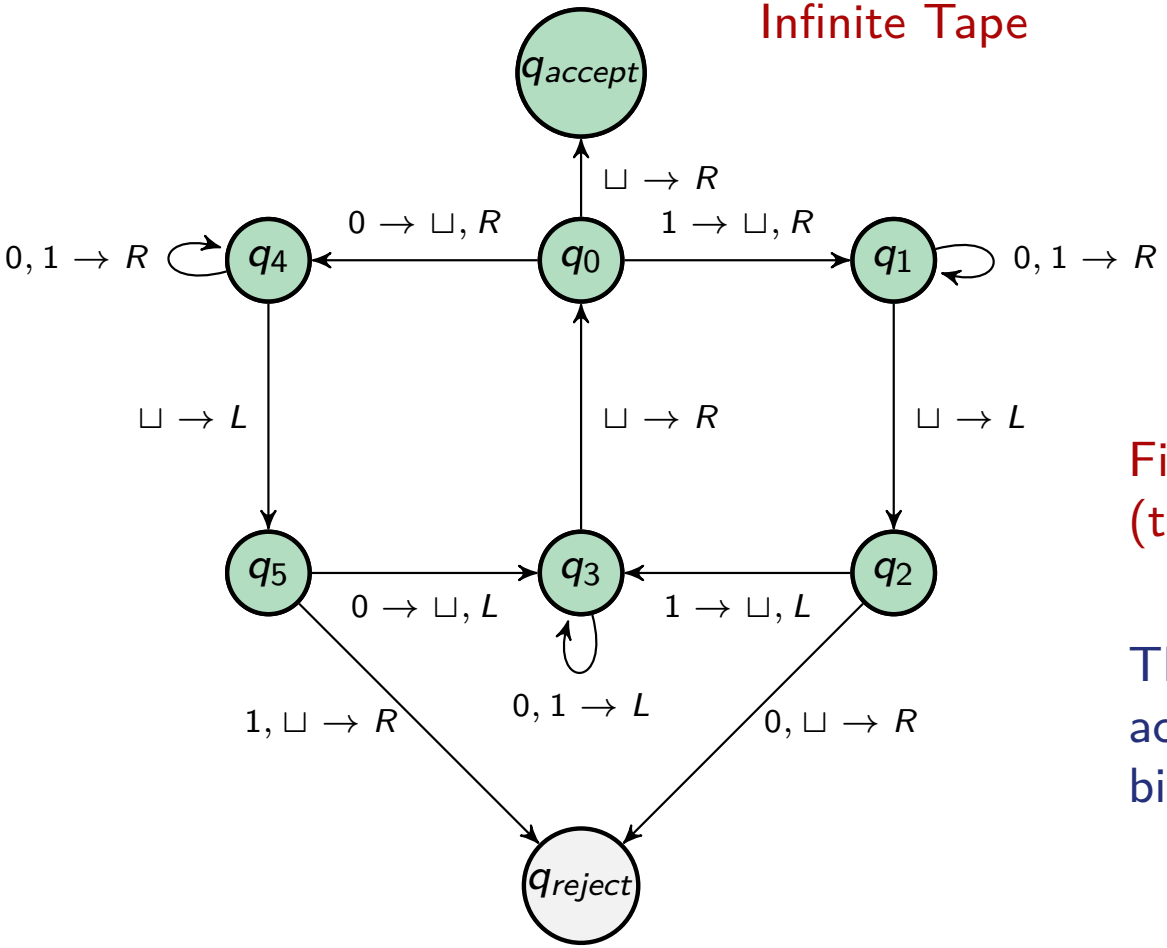
Lecture 5


GREAT. A WAREHOUSE FILLED WITH MILES AND MILES OF REWRITABLE TAPE! WHAT ARE WE EVER GOING TO DO WITH THIS, ALAN?
...ALAN?
And thus the Turing Machine was born.

Recall

The Turing Machine

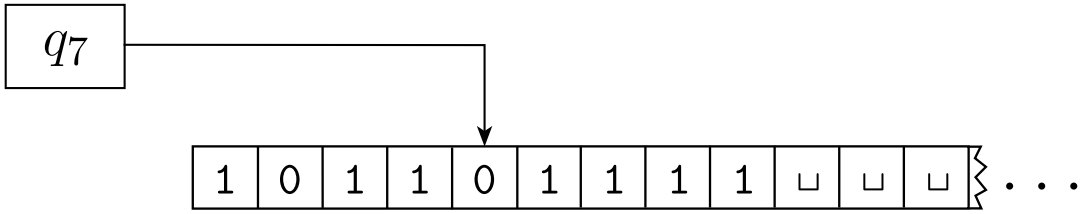




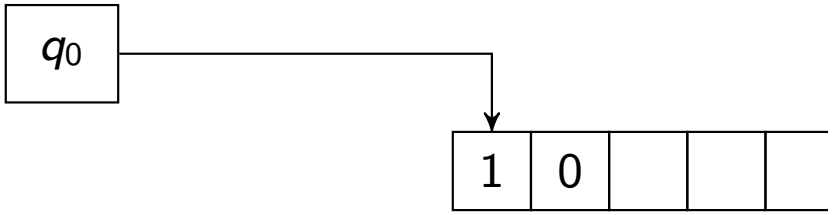
Finite state control (the algorithm)

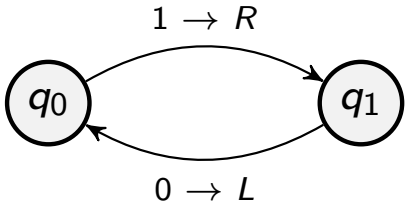
This Turing machine accepts even length binary palindromes

Turing Machine



- Infinite tape
- Tape alphabet contains input alphabet plus ⊔ (blank symbol) plus maybe more symbols
- Head has states (corresponding to the finite control automata)
- Exactly **one** Accept state and exactly **one** Reject state *(where computation immediately ends)*
- Remaining states *"computation in progress"*
- May never reach an accept state. May never halt!





Formal Definition of a TM

A **Turing Machine** is a 7-tuple $(Q, \Sigma, \Gamma, \delta, q_0, q_{accept}, q_{reject})$, where Q, Σ, Γ are all finite sets and

1. Q is the set of states,
2. Σ is the input alphabet not containing the blank symbol $\sqcup$,
3. Γ is the tape alphabet, where $\sqcup \in \Gamma$ and $\Sigma \subseteq \Gamma$,
4. $\delta : Q \times \Gamma \to Q \times \Gamma \times \{L, R\}$ is the transition function,
5. $q_0 \in Q$ is the start state,
6. $q_{accept} \in Q$ is the accept state, and
7. $q_{reject} \in Q$ is the reject state, where $q_{accept} \neq q_{reject}$.

Turing-Recognizable/Decidable Languages

A TM machine M recognizes a language $L \subseteq \Sigma^*$ iff for all inputs $w \in \Sigma^*$:

1. If $w \in L$ then M accepts w and
2. If $w \notin L$ then M either rejects w or never halts

Such languages are called (Turing)-Recognizable

A TM machine M decides a language $L \subseteq \Sigma^*$ iff for all inputs $w \in \Sigma^*$:

1. M halts on w, and
2. M accepts w iff $w \in L$

Such languages are called (Turing)-Decidable

Church–Turing Thesis

Intuitive notion of algorithms	equals	*Turing machine algorithms*

- All algorithms we know of can be executed on TMs
- Anything you write in C, Java, Scala, Python and so on
- The definition is also robust to variations: if we allow for many tapes instead of one, then nothing changes

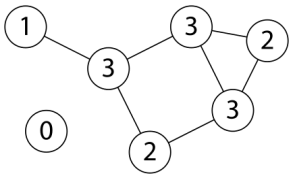
To show language decidable or recognizable it sufficient to describe an algorithm in a high level language (Since by above any such algorithm can be run on a TM)

Decidable Languages

What can TMs compute?

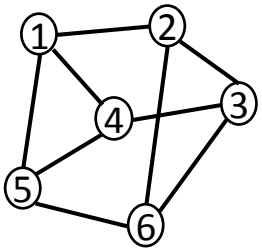
001011 → TM_L → Accept

Equal number of 1's and 0's

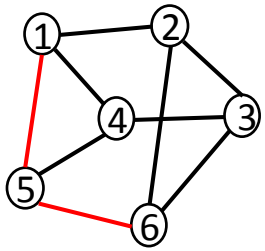


→ $TM_{connected}$ → Reject

Checking connectedness



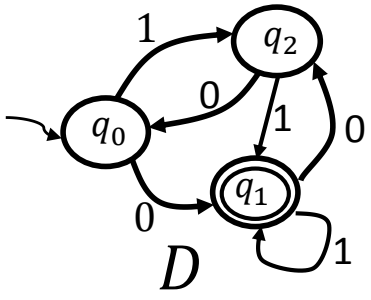
1,6 → $TM_{Shortestpath}$ →



Shortest path problem

And more…

How to encode inputs



Σ	Q	F	q_0	$*$
q_0	1	q_2	$*$	
q_0	0	q_1	$*$	
q_1	0	q_2	$*$	
$\vdots$	...			

Q	Σ	F	q_0	$*$	q_0	1	q_2	$*$	q_0	0	...

$\langle D \rangle \in \{0, 1\}^\star$:

0	1	...	1	1	...	0	0	1	0	1	1	0	0	0	1	0	...	0	0	1	...

Anything can be encoded using binary strings!

Checking Emptiness of a DFA

$$E_{DFA} = \{\langle D \rangle : L(D) = \emptyset\}$$

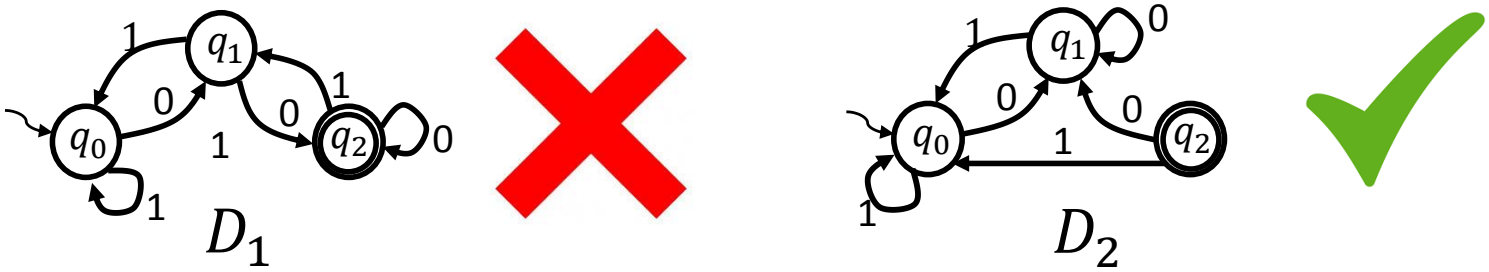
First approach:

1. For each string $s \in \{\varepsilon, 0, 1, 00, 01, 10, 11, 000, \ldots\}$
2. Simulate D on s
   - If D accepts s, THEN *reject*
   - ELSE, pick the next string and go to 2.

If $L(D) = \emptyset$, then this TM will never halt!

Checking Emptiness of a DFA (2nd attempt)

$$E_{DFA} = \{\langle D \rangle : L(D) = \emptyset\}$$



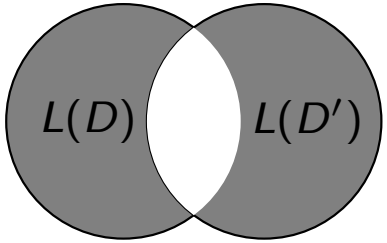
Language accepted by a DFA is non-empty iff there is an accepting state that can be reached from the starting state by a sequence of transitions

Given $\langle D \rangle$ with $D = (Q, \Sigma, \delta, q_0, F)$

1. Initialize $R := \{q_0\}$.
2. Search for $q \in R$, $q' \in Q \setminus R$, $a \in \Sigma$ such that $\delta(q, a) = q'$.
3. If such q, q', a are found then update $R := R \cup \{q'\}$, and go back to Step 2.
4. If none are found then R consists of all states reachable from q_0.
5. Accept iff $R \cap F = \emptyset$

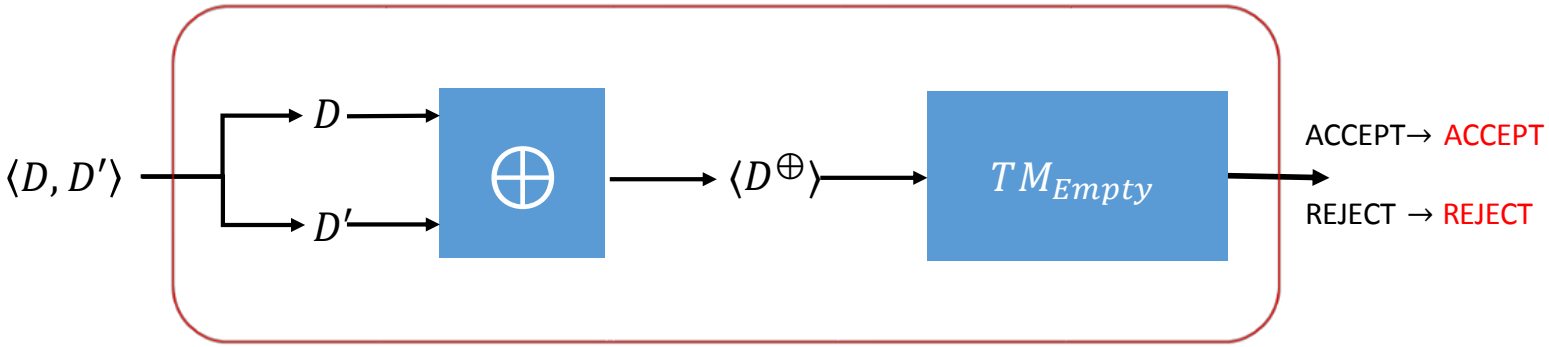
Two DFAs recognize the same language?

$$EQ_{DFA} = \{\langle D, D' \rangle : L(D) = L(D')\}$$



Fact: $L(D) = L(D')$ if and only if $L(D) \oplus L(D') = \emptyset$

- Here $L(D) \oplus L(D') := \left(L(D) \cap \overline{L(D')}\right) \cup \left(\overline{L(D)} \cap L(D')\right)$
- Thus $L(D) \oplus L(D')$ is regular
- Let $D^{\oplus}$ be DFA accepting $L(D) \oplus L(D')$



TM_{EQ}

We get it: Turing machines can compute a lot

But can they compute everything? NO

Theorem (Turing 1936): Halting problem

$$HALT = \{\langle M, w\rangle : M \text{ is a TM and } M \text{ halts on input } w\}$$

is undecidable

Intuitively the difficulty is to conclude in a finite number of steps whether M loops forever or is just slow to halt ...

Halting problem is recognizable

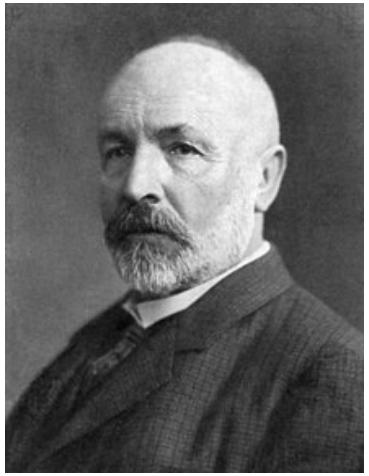
$$HALT = \{\langle M, w\rangle : M \text{ is a TM and } M \text{ halts on input } w\}$$

$HALT$ is recognizable! The following TM U recognizes $HALT$

$U =$ On input $\langle M, w\rangle$ where M is a TM and w is a string:

1. Simulate M on input w.
2. If M ever halts (enters accept/reject state), *accept*.

- Note that U loops on input $\langle M, w\rangle$ if M loops on w, which is why this machine does not decide $HALT$
- Since we will show that $HALT$ is undecidable this shows that recognizers *are* more powerful than deciders



George Cantor (1845–1918)

Diagonalization

Comparing the size of infinite sets

In 1873, Cantor thought about the size of infinite sets

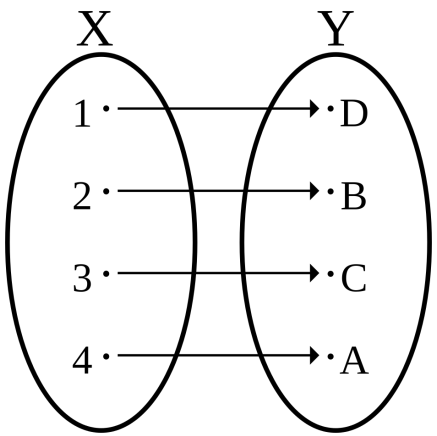
- Is the set of natural numbers $\mathbb{N} = \{1, 2, 3, \ldots, \}$ of the same size as the set of even numbers $\{2, 4, 6, \ldots, \ldots\}$?
- Is the size of $\mathbb{N}$ equal to the size of the set of rational numbers $\mathbb{Q} = \{\frac{m}{n} : m, n \in \mathbb{N}\}$?
- Is the size of $\mathbb{N}$ equal to the size of the set of real numbers $\mathbb{R}$?

Comparing the size of infinite sets

Recall that a function $f : X \to Y$ is bijective if it is

injective it never maps two different elements to the same place – that is, $f(a) \neq f(a')$ whenever $a \neq a'$

surjective it hits every element of Y – that is, for every $b \in Y$ there is an $a \in X$ such that $f(a) = b$



Definition. A set A is countable if either it is finite or it has the same size as $\mathbb{N}$ (i.e., there is a bijection between A and $\mathbb{N}$)

Comparing the size of infinite sets

In 1873, Cantor thought about the size of infinite sets

- Is the set of natural numbers $\mathbb{N} = \{1, 2, 3, \ldots, \}$ of the same size as the set of even numbers $\{2, 4, 6, \ldots, \ldots\}$? YES
- Is the size of $\mathbb{N}$ equal to the size of the set of rational numbers $\mathbb{Q} = \{\frac{m}{n} : m, n \in \mathbb{N}\}$? YES
- Is the size of $\mathbb{N}$ equal to the size of the set of real numbers $\mathbb{R}$? NO

$\mathbb{N}$ and the set of even numbers are of the same size by the bijection $f(n) = 2n$:

n	$f(n)$
1	2
2	4
3	6
$\vdots$	$\vdots$

$\mathbb{N}$ and $\mathbb{Q}$ are of the same size by a bit more complex bijection (see book).

May seem bizzarre at first but incredibly influential!

"No one shall expel us from the paradise which Cantor has created for us."

The reals are uncountable

- Suppose toward contradiction that there is a bijection $f : \mathbb{N} \to \mathbb{R}$
- We reach a contradiction by defining an $x \in \mathbb{R}$ such that no number $a \in \mathbb{N}$ maps to x, i.e., $f(a) = x$
- For every $n \in \mathbb{N}$, the n-th fractional digit of x is selected to be different from the n-th digit of $f(n)$
- Example:

n	$f(n)$
1	$3.\underline{1}4159\ldots$
2	$55.5\underline{5}555\ldots$
3	$0.12\underline{3}45\ldots$
4	$0.500\underline{0}0\ldots$
$\vdots$	$\vdots$

$$x = 0.4641\ldots$$

- By definition there is no $n \in \mathbb{N}$ such that $f(n) = x$ which contradicts that f is onto.

We now use Diagonalization to

- First prove that there is a language that is not decidable
- Then prove that the specific language

$$HALT = \{\langle M, w\rangle : M \text{ is a TM and } M \text{ halts on } w\}$$

is not decidable

There are undecidable languages

- **Turing machines are countable:** enumerate all encodings

	$\langle M_1 \rangle$	$\langle M_2 \rangle$	$\langle M_3 \rangle$	$\langle M_4 \rangle$	$\langle M_5 \rangle$	$\langle M_6 \rangle$	$\dots$
M_1	A	∞	R	A	A	R	$\dots$
M_2	R	R	A	A	∞	A	$\dots$
M_3	R	∞	A	∞	R	R	$\dots$
M_4	A	∞	R	R	R	∞	$\dots$
M_5	∞	∞	A	A	A	A	$\dots$
M_6	R	A	R	∞	A	∞	$\dots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

Let $DIAG = \{\langle M_i \rangle : M_i \text{ doesn't accept } \langle M_i \rangle\} = \{\langle M_2 \rangle, \langle M_4 \rangle, \langle M_6 \rangle, \dots\}$

Theorem *DIAG* is undecidable

Proof by contradiction

- Suppose M decided $DIAG$
- Then $M = M_i$ for some $i \in \mathbb{N}$
- We have $L(M_i) = DIAG$

Does $\langle M_i \rangle \in L(M_i)$? Two cases

- If $\langle M_i \rangle \in L(M_i)$ then by def. of $DIAG$, $\langle M_i \rangle \notin DIAG$ – contradiction
- If $\langle M_i \rangle \notin L(M_i)$ then by def. of $DIAG$, $\langle M_i \rangle \in DIAG$ – contradiction

Thm: $HALT = \{\langle M, w\rangle : M \text{ is a TM and } M \text{ halts on } w\}$ is undecidable

Proof by contradiction

Assume on the contrary that H is a decider for $HALT$

We construct a decider D for $DIAG = \{\langle M\rangle : M \text{ doesn't accept } \langle M\rangle\}$

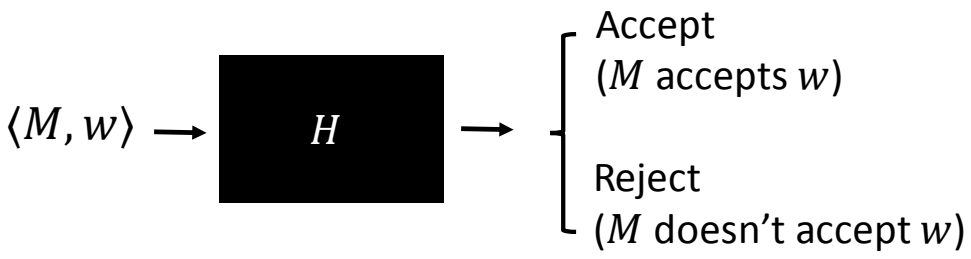
- D: On input $\langle M\rangle$, run H on input $\langle M, \langle M\rangle\rangle$
  1. If H rejects (i.e., M loops on $\langle M\rangle$), *accept*
  2. If H accepts (i.e., M halts on $\langle M\rangle$), run M on input $\langle M\rangle$. When M accepts/rejects, output the opposite.

Need to prove that D decides $DIAG$:

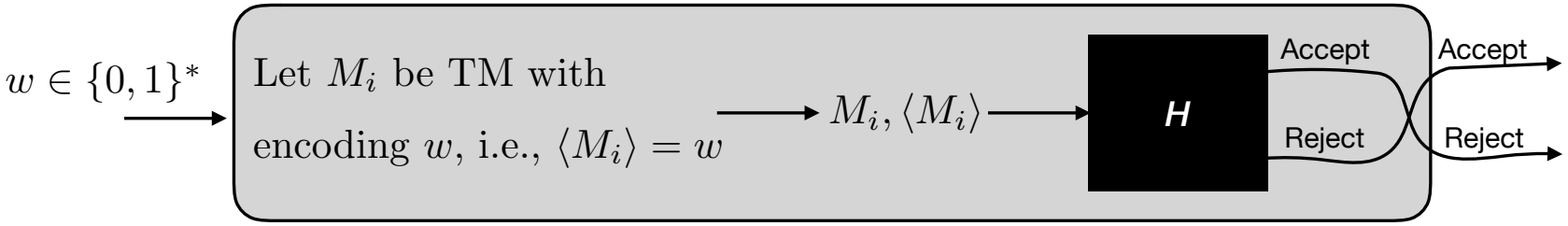
1. D halts on all inputs
2. D accepts $\langle M\rangle \iff M$ does not accept $\langle M\rangle \iff \langle M\rangle \in DIAG$

Thm: $A_{TM} = \{\langle M, w\rangle : M \text{ is a TM and } M \text{ accepts } w\}$ is undecidable

Proof by contradiction Assume on the contrary that H is a decider for A_{TM}

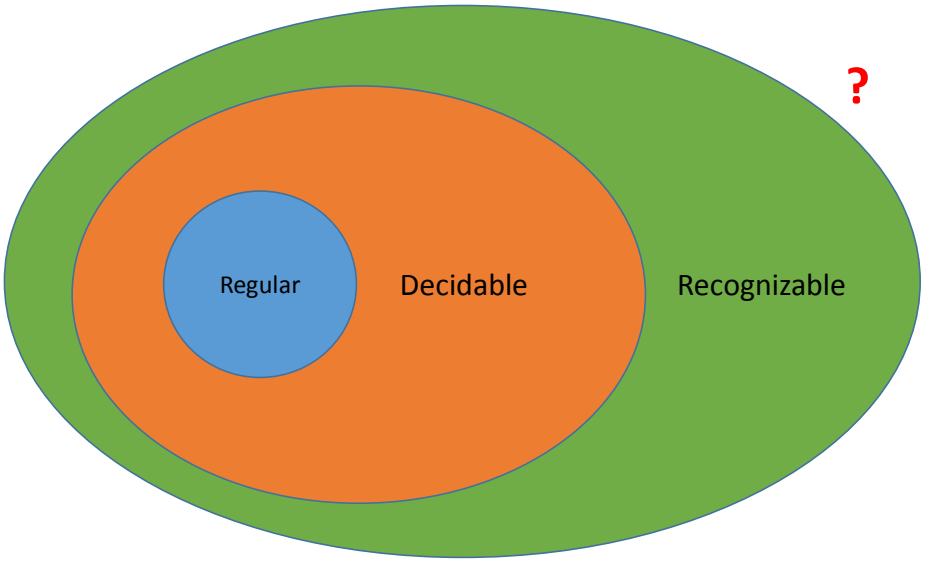


We construct a decider D for $DIAG = \{\langle M_i\rangle : M \text{ doesn't accept } \langle M_i\rangle\}$ using H:



Need to prove that D decides *DIAG*:

1. D halts on all inputs
2. D accepts $\langle M\rangle \iff M$ does not accept $\langle M\rangle \iff \langle M\rangle \in DIAG$



Unrecognizable languages?

Unrecognizable languages exist!

Thm: $\overline{HALT}$ is not recognizable

Step 1

Thm: A language L is decidable iff it is recognizable and its complement is also recognizable

Proof:

- Easy direction: If L is decidable $\Rightarrow L, \overline{L}$ are recognizable
- Harder direction: Assume $L, \overline{L}$ are recognizable
- Let M_1, M_2 be the corresponding recognizers

$\boldsymbol{M}$ = "On input **w**:
1- Run **both** $\boldsymbol{M_1}$ and $\boldsymbol{M_2}$ on input $\boldsymbol{w}$, **in parallel**
2- If $\boldsymbol{M_1}$ accepts, ***accept***; if $\boldsymbol{M_2}$ accepts, ***reject***"

Formally: we alternate between running 1 step of $\boldsymbol{M_1}$ and 1 step of $\boldsymbol{M_2}$

- M decides L
  - For all w, either $w \in L$ or $w \in \overline{L}$
  - Either M_1 accepts w or M_2 accepts w
  - M halts once one of them stops $\Rightarrow$ M must always halt!

Step 2

Corollary: Language $\overline{HALT}$ is unrecognizable

- Recall that
  - $HALT$ is undecidable
  - $HALT$ is recognizable
- So by previous Thm we must have that $\overline{HALT}$ is unrecognizable

Next week: **Reductions**